Vanderbilt University Motorsports
Vanderbilt University | VUM-27 | 2027 Season

HOW-TO GUIDE

GENERAL / ONBOARDING

Overleaf and LaTeX From Zero

Everything you need to write your first VUM document, assuming you have never seen either one

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Summary

Overleaf is the website where we write team documents, and LaTeX is the language you type into it. No prior experience with either is assumed, and both are less complicated than they first appear. Work through this guide once from top to bottom with a laptop open. By the end you will have written your member introduction, the first of the two onboarding tasks. Budget 45 min.

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A	2026-08-13	D. Lu	First release, written for the 2027 intake.
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Master copy: https://github.com/vanderbilt-motorsports/documentation

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1 Overview

Every technical document this team produces, from design reports to test procedures to guides like this one, is written in Overleaf. Two terms need to be distinguished before anything else:

Overleaf is a *website* where the writing happens. Like Google Docs, it runs in your browser, saves automatically, and supports several people editing the same document at once.

LaTeX (pronounced "LAY-tek" or "LAH-tek") is the *language* you type. Instead of clicking a Bold button, you type an instruction meaning "make this bold", and Overleaf turns the full set of instructions into a finished PDF.

In practice, you type instructions in the left half of the screen and the finished PDF appears in the right half.

1.1 Reasons for Using LaTeX

For a single short document, Word would probably be quicker. We use LaTeX because our documents have to remain usable long after the people who wrote them have graduated, and it has several properties that support that:

- **Every document looks the same.** Formatting lives in one shared file rather than in each author's habits, so a report cannot accidentally look different from the rest of the archive.
- **Equations and units are typeset correctly** without a separate equation editor.
- **Figures, tables, and references number themselves.** Inserting a figure into the middle of a finished report renumbers every reference to it automatically.
- **Documents remain portable.** A LaTeX source file produces the same output on any machine in any year, which is not true of a Word file with embedded objects.

Key Takeaway

This guide teaches the twenty or so commands our template provides rather than LaTeX in general. The template handles the difficult parts.

Background

This team has lost roughly two decades of know-how because it was not written down. Test setups left with the people who knew them, results stayed behind in group chats, and some fixes were worked out more than once. The documentation system exists to rebuild that knowledge, and everything written in it remains with the team after its author graduates.

2 The Onboarding Tasks

Onboarding consists of two tasks, which together cover the two tools the whole team runs on while nothing yet depends on the result:

Step 1. **Write your member introduction** in Overleaf, a one-page document about who you are and what you want to work on. This guide covers that task.

Step 2. **Push it to GitHub**, to the `MemberIntroductions` repository. `GEN-HG-001` walks you through that half.

Completing both tasks produces a real document in the house format and your first commit, and every later document and commit follows the same procedure.

> **Content guidelines for the introduction**
>
> The introductions repository is **private**, but everyone in the GitHub organisation can still read it, including members who have not joined yet, and git keeps things effectively forever, since deleting a file later does not remove it from the history.
> So pitch it the way you would introduce yourself at the first team meeting. Your name, year and major, what you want to work on, what you can already do, what you want to learn, something non-car about you. **Do not include** your home address, phone number, student ID or VUnetID, date of birth, grades, class schedule, health or financial information, or anything about somebody else who has not agreed to it. A photo is optional.

3 Account Setup

Vanderbilt provides Overleaf Professional to students, so there is no need to use the free tier or to pay for an individual subscription.

Step 1. Go to `https://www.overleaf.com/edu/vanderbilt`.

Step 2. Sign up with your `@vanderbilt.edu` address. If you already have an Overleaf account under a personal email, sign in and add your Vanderbilt address under **Account Settings** → **Emails and Affiliations** → **Add another email**, then confirm it from your inbox.

Step 3. Confirm the affiliation when prompted. Your account should then show **Professional** rather than Free.

If the account still shows Free, the confirmation email has usually not been opened yet.

Step 4. Ask a lead to share the template project with you.

For leads, the one-time setup of uploading and sharing the template is described in `README.md`, and it assumes as little prior knowledge as this guide does. The system is new to the whole team, leads included.

Existing paid subscriptions

Link your Vanderbilt email first, then email `support@overleaf.com` to ask for a prorated refund, which they handle routinely.

4 The Editor Layout

Opening any project shows a single screen with three columns. Before changing anything, spend a minute matching Figure 1 to what is on your monitor.

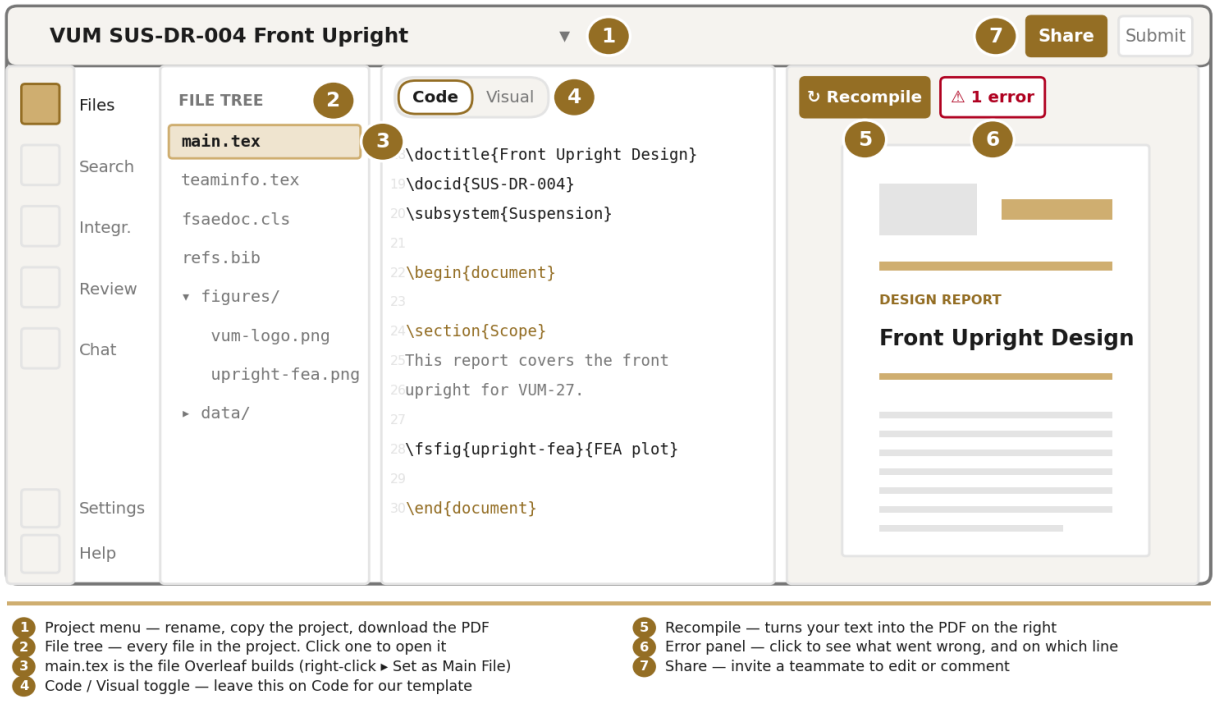


Figure 1. The Overleaf editor. Your files on the left, what you type in the middle, the finished PDF on the right.

Only two controls matter at first: **Recompile**, which turns your text into the PDF, and the **file tree**, which opens a different file.

Leave the editor on "Code"

Overleaf also offers a **Visual** editor that hides the instructions and behaves more like Word. It does not understand our template's commands and renders them as grey placeholders that cannot be edited properly, so keep the toggle on **Code**.

5 Creating a Document from the Template

Do not edit the template project itself. Copy it and work in the copy.

Step 1. Open the template project.

Step 2. Click the dropdown next to the project name at the top and choose **Copy Project**.

Step 3. Name the copy after the document you are writing, using its document ID. For your introduction that is `VUM MI-2026-LU Member Intro`, and for a later report it might be `VUM SUS-DR-004 Front Upright`.

Avoid names like "report" or "my project", because the project has to be findable years from now.

Step 4. In the copy, decide which starter file you need. For your introduction that is `member-intro.tex`, and later it will be `design-report.tex`, `test-procedure.tex`, `test-report.tex`, or `how-to-guide.tex`. Delete the ones you are not using with right-click → **Delete**.

Step 5. Rename the one you kept to `main.tex` with right-click → **Rename**.

Step 6. Right-click `main.tex` → **Set as Main File**. This tells Overleaf which file to build.

If this step is skipped, Overleaf may build a different file, and edits will appear to have no effect.

Step 7. Press **Recompile**. A PDF containing the example content confirms that the project is set up correctly.

6 The Structure of a `.tex` File

Open `main.tex`. Every document consists of two parts.

```
% Anything after a percent sign is a comment. LaTeX ignores it.

\documentclass[designreport]{fsaedoc}   %  <-- PART 1: the preamble
\input{teaminfo}                        %      starts at the top and runs
                                        %      to \begin{document}
\doctitle{Front Upright Design}         %      This is where you put the
\docid{SUS-DR-004}                      %      title, ID, authors and
\subsystem{Suspension}                  %      revision history.

\begin{document}                        %  <-- PART 2: the body
                                        %      everything between
\section{Scope}                         %      \begin{document} and
                                        %      \end{document} is what
This report covers the front upright.   %      actually gets printed.

\end{document}                          %      Nothing after this exists.
```

In our starters the preamble is fenced off with a comment banner that says `FILL THIS IN`. Complete every field in that block before writing the body, which takes about two minutes and is easy to overlook.

Table 1. The two halves of every document

Part	What goes there
Preamble	Facts *about* the document: title, ID, subsystem, authors, revision history. Settings, not content.
Body	The document itself: your sections, your writing, your figures and tables.

> **Preamble commands in the body**
>
> If a preamble command such as `\doctitle{...}` is placed *below* `\begin{document}`, the build stops with the error *"Can be used only in preamble"*. Nothing is damaged. The line is in the wrong half of the file and needs to move up.

7 The Five Rules of Typing in LaTeX

These five rules cover all the LaTeX you need for our documents.

7.1 A backslash starts a command

Anything beginning with `\` is an instruction rather than text. For example, `\section` makes a heading and `\textbf` makes bold.

7.2 Curly braces hold what the command acts on

```
\section{Scope and Objectives}
\textbf{this text comes out bold}
```

Some commands take several arguments in braces, one after another. The order matters, and the cheat sheet lists it for each command:

```
\fsfig{upright-fea}{Von Mises stress under the 3 g bump case.}
%       ^ which image        ^ what the caption says
```

7.3 Square brackets are optional extras

If a command has square brackets in front of the braces, that part is optional and has a sensible default, so it can be left out entirely, as in the following pair:

```
\fsfig{upright-fea}{A picture.}        % default width
\fsfig[0.5]{upright-fea}{A picture.}   % half as wide
```

7.4 Environments come in `begin` and `end` pairs

When an instruction applies to a whole block rather than a few words, it gets a `\begin` and a matching `\end`. These pairs are called **environments**, as in the following list:

```
\begin{itemize}
  \item first bullet
  \item second bullet
\end{itemize}
```

> **Tip**
>
> Type the `\end` line immediately after the `\begin` line, then go back and fill in the middle. Unclosed environments produce the most confusing error messages, and this habit prevents them.

7.5 A blank line starts a new paragraph

This behaviour differs from every word processor.

```
These two lines
end up on the same line in the PDF.

This one starts a new paragraph, because of the blank line above it.
```

Pressing Enter once has no visible effect, while pressing it twice starts a new paragraph. Ten consecutive spaces are likewise printed as a single space, so spacing cannot be adjusted by hand. The template controls the layout.

7.6 Reserved characters

Seven characters have special meaning to LaTeX, so printing them literally requires a backslash in front:

Table 2. Reserved characters

You want	You type	Why
%	\%	starts a comment
&	\&	separates table columns
_	_	makes a subscript
#	\#	marks a command's argument
$	\$	switches maths on and off
{ }	\{ \}	group things together

So `R&D at 100%` must be typed `R\&D at 100\%`. If the backslash before a `%` is forgotten, the rest of the line is treated as a comment and disappears from the PDF without an error.

> **Maths**
>
> Anything between dollar signs is set as maths: `$F_z$` gives F_z, and `$\sigma_{vm}$` gives σ_{vm}. This is why a bare underscore is reserved, because outside dollar signs LaTeX cannot interpret it.

8 Compiling and Reading Errors

Press **Recompile**. One of three things happens, as summarised in Table 3.

Table 3. Possible outcomes of a recompile

Result	What it means
A PDF appears	The build succeeded.
Yellow “N warnings”	The build succeeded with minor issues, which are usually safe to ignore while drafting.
Red “N errors”	The build failed, and the PDF is unchanged until the error is fixed.

When a red error appears, work through it as follows, and refer to Table 4 for the common cases.

Step 1. Click the red button next to **Recompile**.

Step 2. Read only the **first** error. Everything after it is usually the same mistake echoing.

A single missing brace can generate fifteen errors, and fixing the first usually clears the rest.

Step 3. Click the line number in the error. Overleaf jumps your cursor to it.

Step 4. Look at that line **and the one above it**. LaTeX often only notices a problem on the line after the one that caused it.

Table 4. Common errors and their causes

Message	What is really wrong
`Undefined control sequence`	A command is misspelled, or does not exist. Check the cheat sheet.
`Missing $ inserted`	An unescaped `_` or `^` outside maths. Wrap it in dollar signs.
`Can be used only in preamble`	A setup line ended up below `\begin{document}`. Move it up.
`Missing } inserted`	A brace was opened and never closed.
`Environment ... undefined`	A misspelled `\begin{...}` name.
`File ... not found`	An image name is wrong, or the file is not in `figures/`.
`Extra alignment tab`	Too many `&` in a table row.
`??` in the PDF	Not an error. Recompile, since cross-references need two passes.

Finding an error by bisection

Comment out half the document by putting `%` at the start of each line (select the lines and press **Ctrl+/** or **Cmd+/**), then recompile. If the error disappears, it is in the hidden half, and otherwise it is in the visible half. Repeating this on the affected half locates any error within a few iterations.

9 Adding a Picture

Step 1. Obtain the image as a `.png`, `.jpg`, or `.pdf`. A screenshot is acceptable.

Step 2. Drag it onto the `figures/` folder in the file tree. Give it a sensible name with no spaces: `upright-fea.png`, not `Screenshot 2026-08-13 at 11.42.07.png`.

Step 3. Put this on its own line where you want it, and note that you write the name **without** the `.png`:

```
\fsfig{upright-fea}{Von Mises stress under the 3 g bump case.}
```

Step 4. Recompile. The figure can then be referenced anywhere in the text with `\reffig{upright-fea}`, and LaTeX fills in the correct number.

Figure placement

LaTeX treats figures as "floats" and places them wherever they fit without leaving a large gap, which is sometimes a later page. References are therefore written as "see `\reffig{...}`" rather than "see the figure below": the number is always correct even when the position moves.

10 Working With Other People

Step 1. **Share** (top right) invites someone by email. Give reviewers **Can edit** if they will fix things, **Can view** if they will only read.

Step 2. Reviewers should use the **Review** icon in the left bar to leave comments and tracked changes rather than silently rewriting your text.

Step 3. Overleaf saves continuously. There is no save button, and closing the tab does not lose work.

Step 4. The project menu has **History**, which shows every version and who wrote what. Earlier versions can be restored from there.

Caution

Two people typing in the same paragraph at the same time works, but two people restructuring the same section simultaneously causes confusion, so say in the team chat which section you are working in.

11 Exporting the PDF

Export with the project menu → **Download PDF**, and rename the file to match the document before sharing it anywhere:

```
SUS-DR-004_RevB_Front-Upright.pdf
```

Then attach it to a release in the documentation repository. The team reads released documents from the repository rather than from Overleaf, so a PDF that never leaves an individual account is effectively unavailable to everyone else.

12 Exercise: Writing Your Introduction

The material above only becomes familiar through use, so work through the checklist on your actual introduction rather than a scratch file. The result is the document you will push in the next task.

- ☐ Sign in at the Vanderbilt Overleaf portal and confirm your affiliation
- ☐ Copy the template project and name the copy `VUM MI-<year>-<SURNAME> Member Intro`
- ☐ Keep `member-intro.tex`, delete the other starters, rename it to `main.tex`, set it as the main file
- ☐ Recompile and confirm you get a PDF of the example content
- ☐ Read the comment banner at the top of the file, which lists what to include and what to leave out
- ☐ Replace `\doctitle{}` with your name and `\docsubtitle{}` with your year and major
- ☐ Set `\docid{}` to `MI-<year you joined>-<SURNAME>` and `\docauthor{}{}` to your name and email
- ☐ Rewrite every section of the body in your own words, deleting any you have nothing to say about and adding any you want
- ☐ Add a bullet list somewhere, so that you have used one once
- ☐ Optionally add a photo: drop it in `figures/` and uncomment the `\fsfig{}{}` line at the bottom
- ☐ Delete a closing brace deliberately, then practise reading the resulting error
- ☐ Fix it, recompile, and check the PDF says what you meant
- ☐ Download the PDF *and* the `main.tex` source, since both are needed for the next task

Downloading the source

The PDF comes from the project menu → **Download PDF**. For the `.tex`, click `main.tex` in the file tree and then the download arrow above the file list, or download the whole project with **Download Source** and take `main.tex` from the zip.

Completing the checklist covers everything this guide sets out to teach. The remaining commands are listed in `CHEATSHEET.md` in the project.

> **Key Takeaway**
>
> GEN-HG-001 continues from here, taking the two downloaded files through the GitHub half of onboarding.

13 Troubleshooting

Step 1. Re-read the **first** error, not the last.

Step 2. Undo (**Ctrl+Z** / **Cmd+Z**) back to the last version that compiled, and redo your change in smaller pieces.

Step 3. Use the comment-out trick from the error section to find the guilty line.

Step 4. Compare against a line in the starter file that already works; copying that line and changing its contents resolves most problems.

Step 5. **After ten minutes, ask.** Post the error message and the affected line in the team chat. A second reader usually finds this class of error quickly.

> **Key Takeaway**
>
> A good VUM document does not require a deep understanding of LaTeX. Fill in the header block, write clearly, and copy the small set of commands you actually use. The effort belongs in the engineering content.

> **Keeping this guide current**
>
> Overleaf redesigns its interface periodically, and the button names here go stale when it does. If something no longer matches the interface, fix this file and add a revision line at the top.

[OPEN ITEM] *Replace the schematic in Figure 1 with real annotated screenshots once we have a screen-capture licence sorted (D. Lu, 2027-02-01)*

Open Items

Owner	Item	Target Date
D. Lu	Replace the schematic in Figure 1 with real annotated screenshots once we have a screen-capture licence sorted	2027-02-01